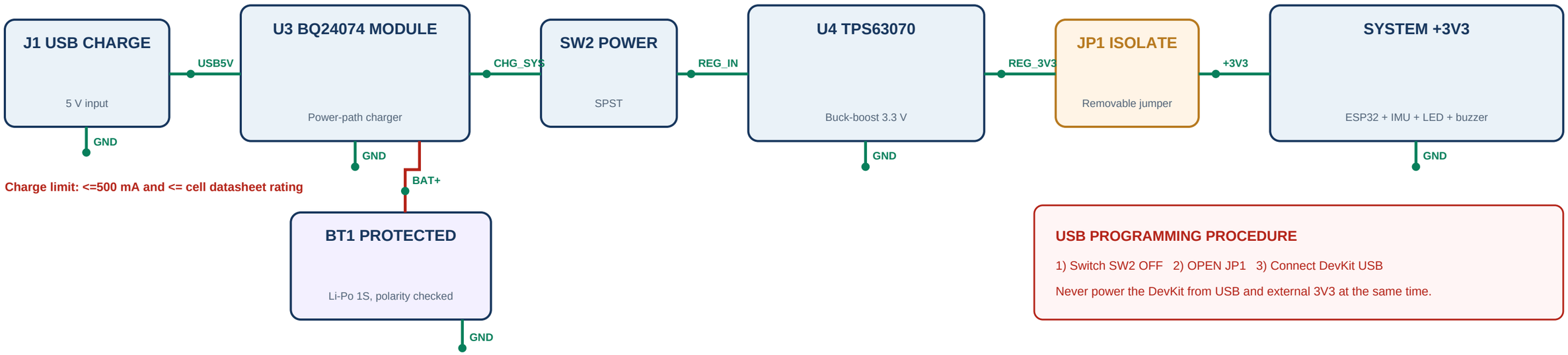
SMART FALL-GUARD - HARDWARE SCHEMATIC

26 OCT 2026

SP1-HW-01 | ESP32 DevKit 30-pin | MPU6050 | USB-first + protected Li-Po power path | Rev B

1. POWER, CHARGING AND USB PROGRAMMING ISOLATION

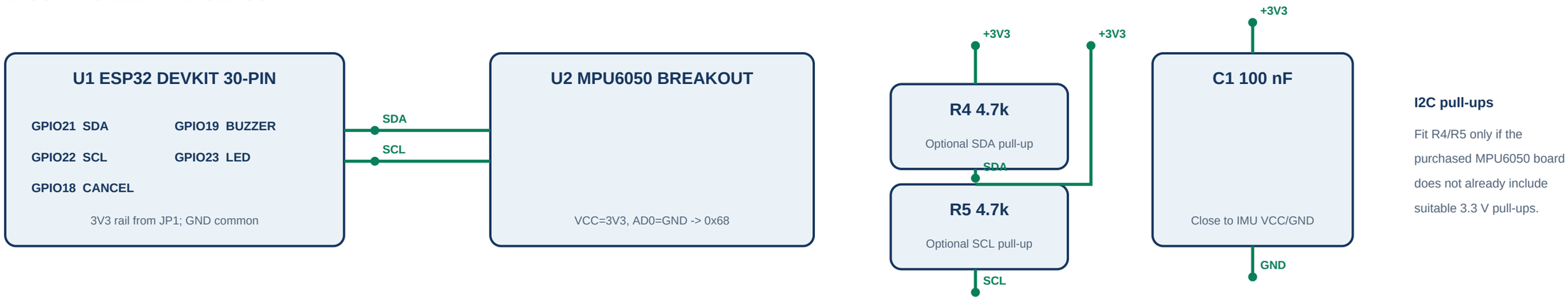


USB PROGRAMMING PROCEDURE

1) Switch SW2 OFF 2) OPEN JP1 3) Connect DevKit USB

Never power the DevKit from USB and external 3V3 at the same time.

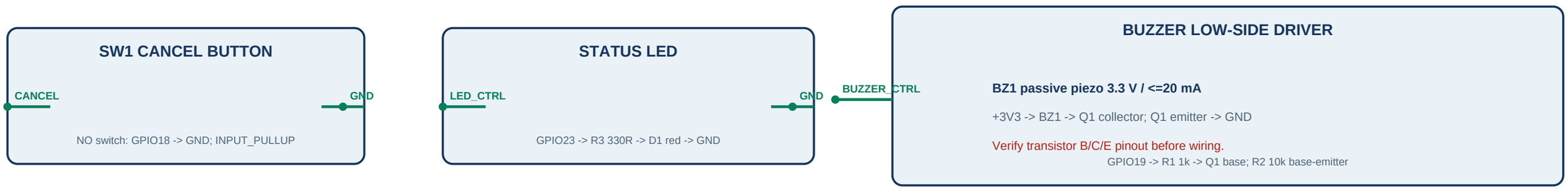
2. CONTROLLER AND SENSOR



I2C pull-ups

Fit R4/R5 only if the purchased MPU6050 board does not already include suitable 3.3 V pull-ups.

3. USER INPUT, STATUS LED AND BUZZER DRIVER



DESIGN STATUS: schematic complete at module level; footprints and physical pinout are intentionally pending purchased-part verification.